

Alex M. Cooper Jr.

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PROFESSIONAL SUMMARY

Computer Science graduate with a 3.79 GPA and hands-on experience in systems administration, cross-platform environment management, and IT troubleshooting. Holds certificates in Cloud Engineering and Cyber Security with experience managing Linux and Windows servers, deploying containerized applications, and resolving hardware and software issues across enterprise and research settings.

EXPERIENCE

LUCY Mission Intern, Systems Engineering

June 2026 – Current

National Aeronautics and Space Administration (NASA)

Greenbelt, Maryland

- Managed directories for licenses, proprietary files, and documentation across remote Linux servers, Windows environments, and Mac set-ups.
- Consolidated 3 legacy script files into 1, reducing required command-line executions from 4 to 2, cutting execution steps by 50%.
- Investigated compatibility requirements and communicated needs to enable legacy peripherals to function within modern computing environments.

L'SPACE Lead Systems Engineer

January 2026 – May 2026

National Aeronautics and Space Administration (NASA)

Remote

- Led a team of eight engineers as Lead Systems Engineer, providing logistical and technical support for mission task solution research and design.
- Designed a Lunar Surface vehicle engineering solution and authored a Preliminary Design Review proposal, presenting to technical and non-technical audiences.
- Earned L'SPACE skill badges in System Topics, Project Management, and NX CAD.

Capstone Hardware/Integration Engineer

August 2025 – December 2025

Edge Optimized Image Codec

West Chester, Pennsylvania

- Applied industry-standard dependency management and containerization tools, including shell scripts and containers, to ensure solution stability across Linux and Windows.
- Designed and implemented a UDP-based data transfer protocol for binary file transmission supporting static and real-time streaming.

Software Engineer

January 2025 – December 2026

Campus Connect

West Chester, Pennsylvania

- Deployed the application on CloudLab, a federal university research environment, with a fully functional CI/CD pipeline using GitHub Actions.
- Built an embedded RAG system enabling an off-the-shelf LLM to answer prompts outside its training data with 90% accuracy.
- Monitored application health and security by analyzing traffic and container logs in real time.

Senior Technician

May 2022 – January 2025

Valvoline Instant Oil Change

Lancaster, Pennsylvania

- Processed client transactions through a ticket management POS system, transferring and escalating clients as needed.
- Troubleshooted standard peripherals and coordinated with regional corporate offices to replace and stock keyboards, mice, scanners, and other required items.
- Collaborated with vendors, ISP, and corporate teams to diagnose and resolve software-related issues and technology outages.

EDUCATION

West Chester University of Pennsylvania

West Chester, Pennsylvania

B.S. in Computer Science, Minor in Applied Statistics. Summa Cum Laude | Dean's List: Spring 2023, Spring 2024 - Spring 2026 | Honor Societies: Alpha-Alfa-Alfa, Upsilon-Pi-Epsilon | Certificates: Cloud Engineering, Cyber Security. GPA: 3.79.

May 2026

SKILLS

Systems and Infrastructure: Linux, Windows Server, Active Directory, Oracle VirtualBox, Remote Desktop, Wireshark, gdb

Cloud and DevOps: AWS, CloudLab, Kubernetes, Docker, Terraform, Helm, GitHub

Development and Tools: FastAPI, Node.js, Spring Boot, .NET, Visual Studio Code, JSON

Data and Analytics: SAS, SPSS, Microsoft Power

Professional Skills: Microsoft Office, Communication, Time Management, Client Satisfaction

CERTIFICATIONS

- Google IT Support Certificate (in progress)